

How Capstone Design Project Prepares Students for Career Success

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What are the top best practices for success in entry level mechanical engineers identified by employers?

- A Best Practice is a capability, method, technique, or approach that consistently delivers superior results and is widely accepted as the most effective way to achieve a desired outcome.
- My number one goal is to help you be successful at your FIRST career opportunity
- Write down THREE best practices you think you should have when you graduate.

Top 10 Best Best Practices of an Entry Level MAE by Employers (ChatGPT)

1. Strong CAD and Simulation Skills
2. Effective Communication
3. Team Collaboration
4. Problem Solving Mindset
5. Understanding Manufacturing Processes
6. Project Management Basics – time management & task prioritization
7. Adaptability & Learning Agility – technology and tools evolve
8. Attention to Detail – small errors can lead to big problems
9. Business & Customer Awareness – customer satisfaction matters
10. Professionalism & Work Ethic – builds trust and credibility

Build a Personal Development Plan for this

1. Master CAD and simulation tools – build a portfolio of completed design projects
2. Improve communication skills – take a technical writing course, present designs to peers and ask for feedback, write technical reports
3. Be an effective and reliable team member – work on cross-functional projects, practice active listening and conflict resolution, demonstrate positive peer reviews
4. Sharpen problem-solving skills – solve real-world problems, participate in design competitions, document lessons learned
5. Design with manufacturability in mind – apply design for manufacturability principles to at least one design project

Build a Personal Development Plan for this

6. Learn project management basics – successfully manage a small project from start to finish and track progress using Gantt charts
7. Cultivate adaptability & learning agility – reflect on feedback and adjust habits
8. Minimize errors and improve precision – peer-review others' work and sharpen your eye
9. Align engineering work with business goals and user needs – attend product launch and customer feedback sessions, analyze how design decisions affect cost and usability
10. Build a reputation for reliability & initiative – keep a journal of accomplishments and lessons

Capstone Design does this

- Two semester sequence where students work in teams (of 4) to design, fabricate, test and demonstrate an engineered product for a customer with complete CAD and analysis
- Faculty mentor and “industry” mentor – regular meetings with both
- Projects cover the gamut of MAE technical topics – some interdisciplinary
- Both semesters are Writing in the Discipline (WID) – regular presentations and reports demonstrating effective approaches with lessons learned and peer review
- First Assignment is to make a team charter of professional responsibilities
- Manage a budget to order parts and materials for manufacturing a prototype
- Make and refine Gantt charts for project management
- Provide weekly activity reports with hours worked for visibility and accountability
- Validate and verify the design with testing and demonstration
- Capstone Design Showcase – last Friday of spring semester open to the public – GO!

Projects

- Some Sponsored by Industry
- Some Sponsored by Non-profits
- Some National (and International) Design Competitions
- Some Support Professors' Cutting Edge Research

Let's Highlight Some Projects

We can just use the MAE Capstone Project Website

<https://engineering.gwu.edu/mechanical-aerospace-engineering-senior-design-projects>

Our GOAL is to prepare you for Success in Your Career as an Engineer

Capstone Design brings together everything you have learned to apply to a real-world project so you can hone your capabilities to be prepared to make an impact on DAY 1 of your career

Happy to Chat

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